Descriptive Statistics

I. Population Statistics
- Central Tendency
- Dispersion
- Normal Distribution

II. Estimates from a Limited Sample
- Sample Statistics & Data Display
- How Good are These Estimates?
- Standard Error of the Mean

III. Classification of Data
- Interval Data
- Categorical Data

IV. BME 423 Student Data Summary

Outline

I. Introduction

- Differences Between R & RStudio
- Installing R & RStudio
- Mandatory Commands
- Setting up/changing working directory
- Help Window

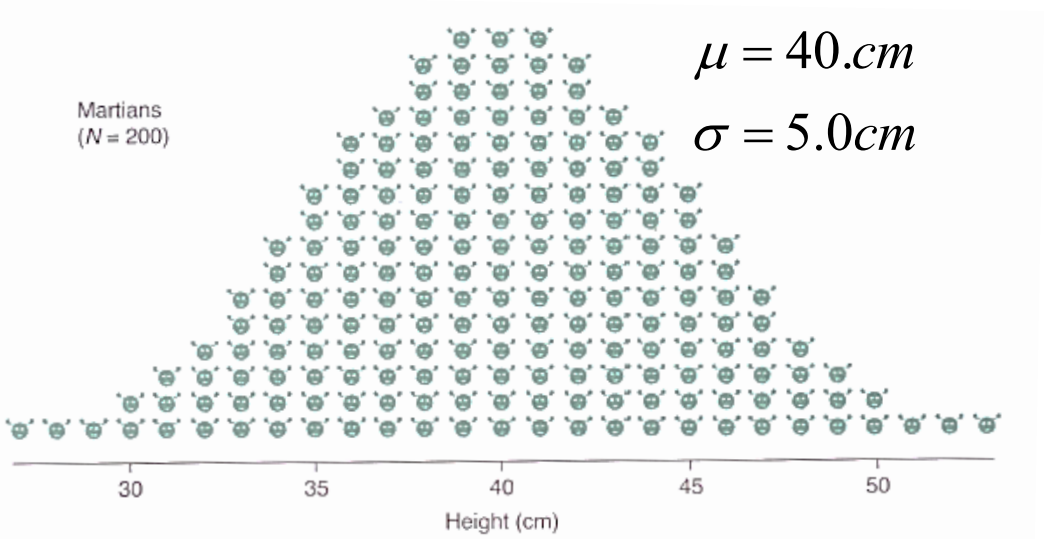
II. Basics

- Variables, vectors, matrices, arrays

III. Data Frame

IV. Installing & loading packages

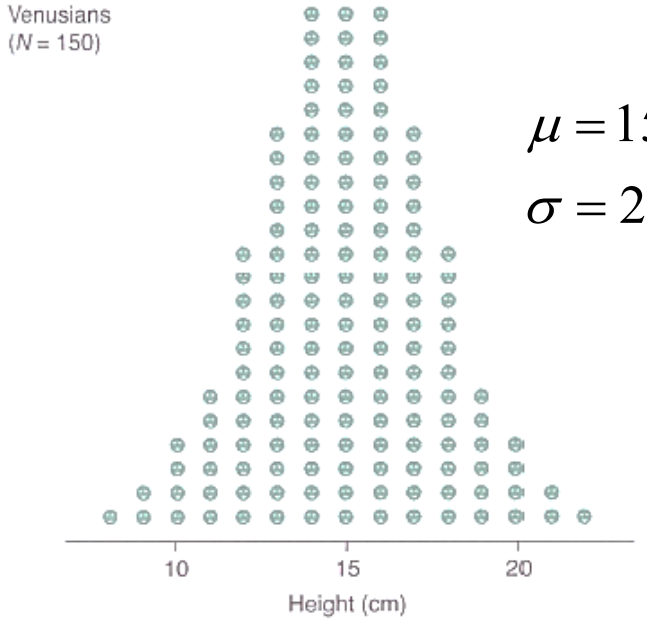
I. Population Statistics



$\mu = 40.cm$

$\sigma = 5.0cm$

FIGURE 2-1. Distribution of heights of 200 Martians, with each Martian's height represented by a single point. Notice that any individual Martian is more likely to have a height near the mean height of the population (40 cm) than far from it and is equally likely to be shorter or taller than average.



$\mu = 15cm$

$\sigma = 2.5cm$

FIGURE 2-2. Distribution of heights of 150 Venusians. Notice that although the average height and dispersion of heights about the mean differ from those of Martians (Fig. 2-1), they both have a similar bell-shaped appearance.

$$x_i \ , \ i = 1, \ldots, \ N$$

Central Tendency - Mean

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Dispersion – Variance $\left(\sigma^2\right)$ or Standard Deviation $\left(\sigma\right)$

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - \mu\right)^2$$

Normally distributed random variable

μ – mean

σ – standard deviation

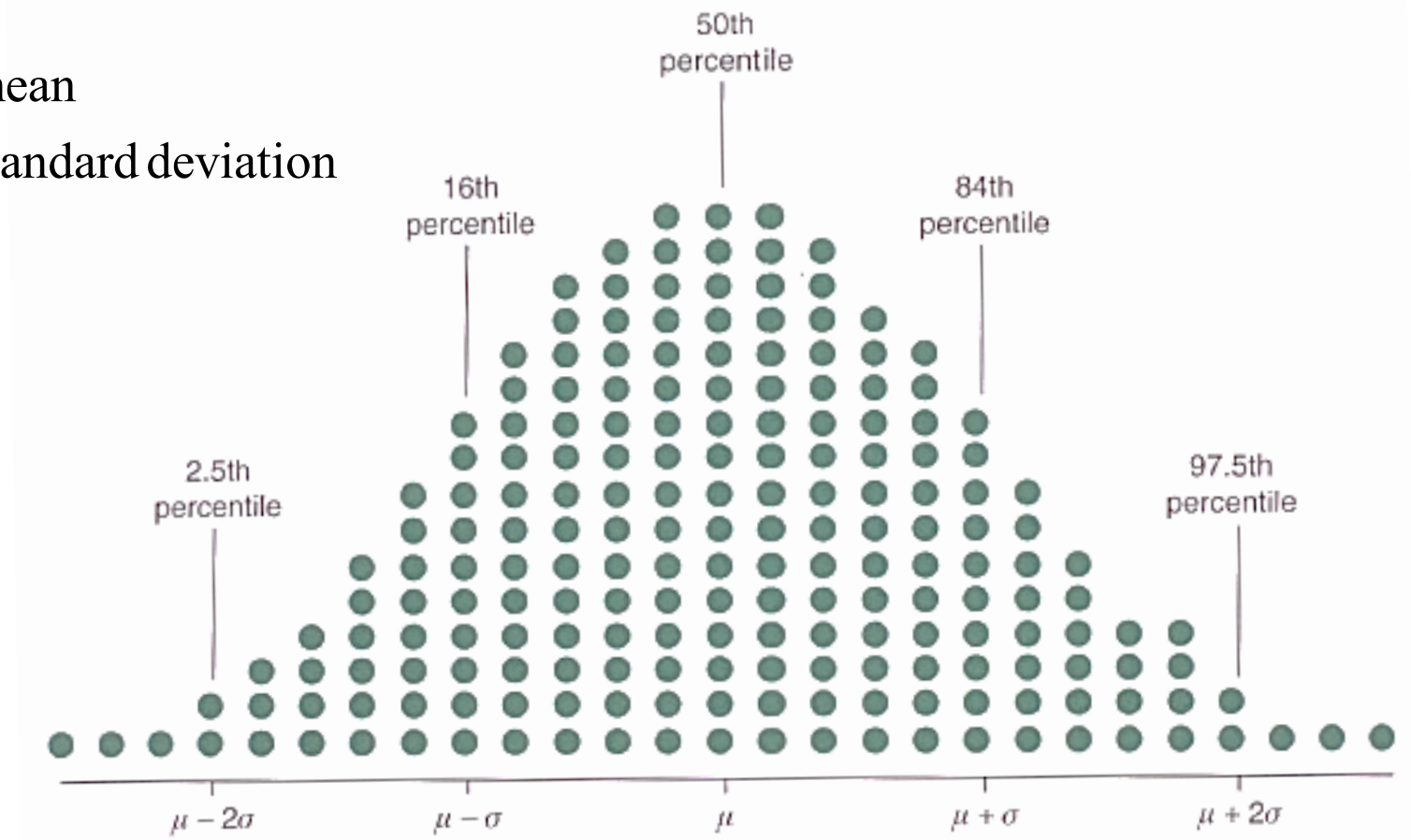


FIGURE 2-10. Percentile points of the normal distribution.

- The proper application of many of the statistical tests for inference introduced in this class will require that the data come from a population that follows a Normal distribution
- The idea of a distribution of a random quantity (including a Normal variable) will be introduced formally when we discuss probability
- For now, several empirical observations about Normal variables will be helpful

Empirical observations about Normal variables

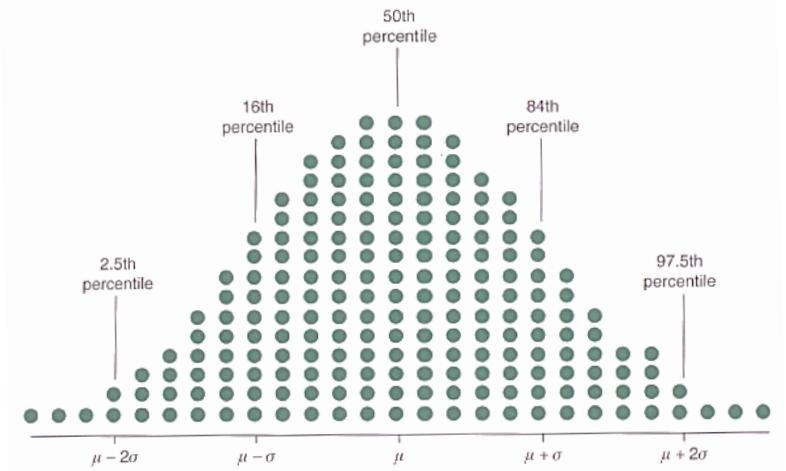


FIGURE 2-10. Percentile points of the normal distribution.

Assume a variable (call it X) follows a Normal distribution with

mean – μ

standard deviation - σ

~2.5% of the values of X will be below the value $\mu - 2\sigma$

~16% of the values of X will be below the value $\mu - \sigma$

50% of the values of X will be below the value μ

~84% of the values of X will be below the value $\mu + \sigma$

~97.5% of the values of X will be below the value $\mu + 2\sigma$

Also:

~68% of the values of X will be between $\mu - \sigma$ & $\mu + \sigma$
(within 1 standard deviation of the mean)

~95% of the values of X will be between $\mu - 2\sigma$ & $\mu + 2\sigma$
(within 2 standard deviations of the mean)

~99% of the values of X will be between $\mu - 3\sigma$ & $\mu + 3\sigma$
(within 3 standard deviations of the mean)

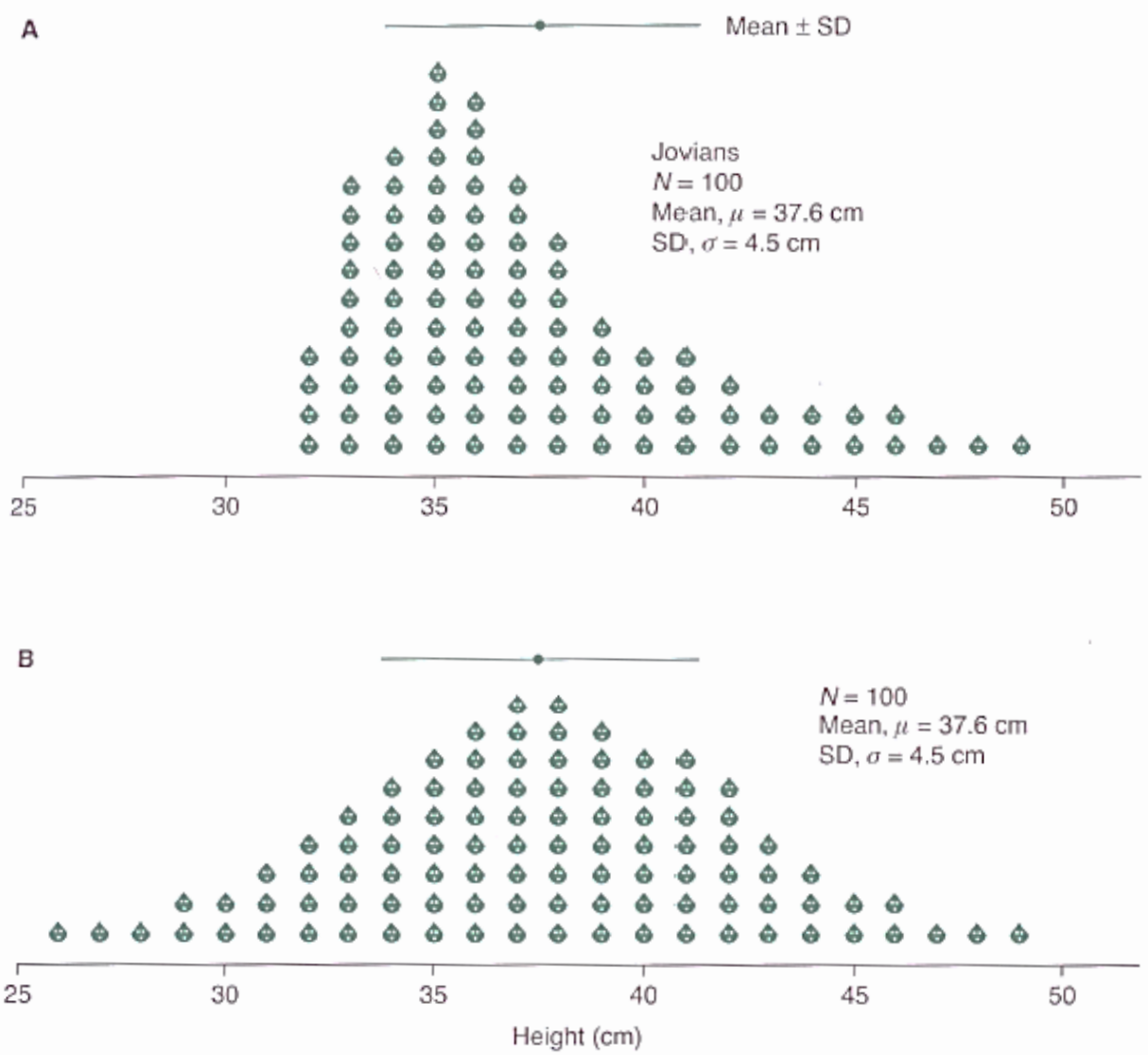


FIGURE 2-8. When the population values are not distributed symmetrically about the mean, reporting the mean and standard deviation can give the reader an inaccurate impression of the distribution of values in the population. Panel **A** shows the true distribution of the heights of the 100 Jovians (note that it is skewed toward taller heights). Panel **B** shows a normally distributed population with 100 members and the same mean and standard deviation as in panel **A**. Despite the fact that the means and standard deviations are the same, the distributions of heights in the two populations are quite different.

Non-Normal Distribution

Additional Information:

- Skewness: measure of lack of symmetry in distribution
- Kurtosis: measure of whether data are "heavy-tailed" or "light- tailed" versus Normal Distribution

Normal Distribution

II. Estimates from a Limited Sample – Sample Statistics

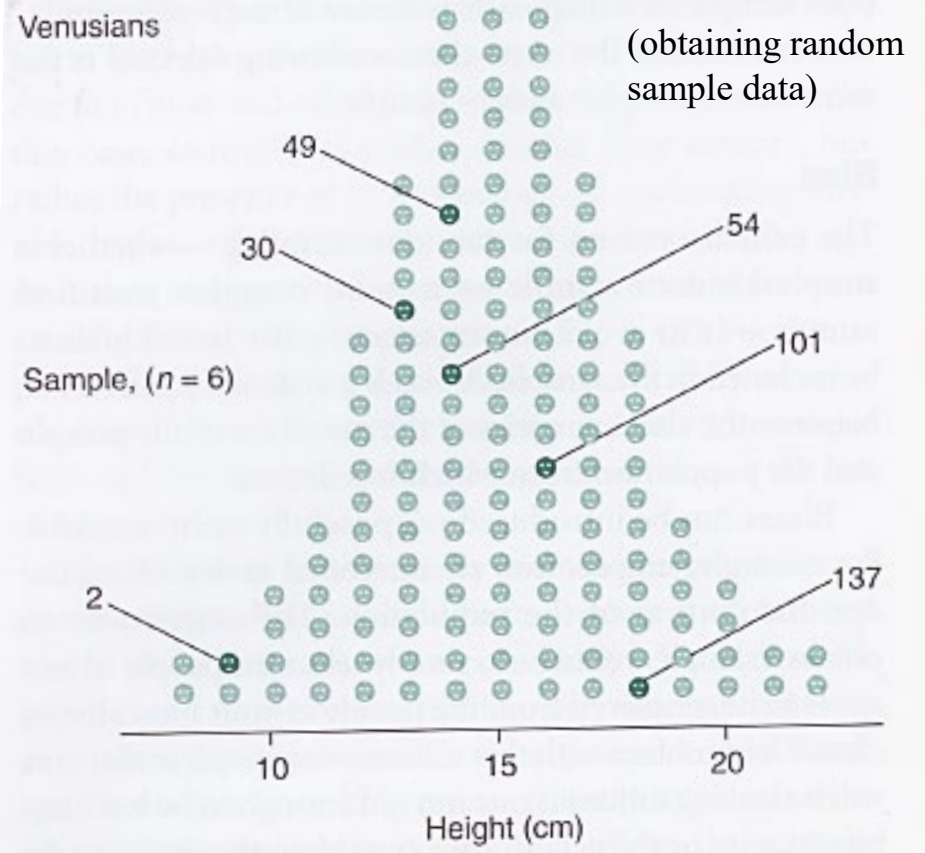


FIGURE 2-3. To select $n = 6$ Venusians at random, we number the entire population of $N = 150$ Venusians from 1 to 150, beginning with the first individual on the far left of the population as number 1. We then select six random numbers from Table 2-2 and select the corresponding individuals for the sample to be observed.

Goal of random sampling is to avoid bias because we want to use sample statistics as estimates for population statistics

$$x_i \ , \ i = 1, \ldots, \ n \qquad n < N$$

Central Tendency – Sample Mean

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Dispersion – Sample Variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2$$

Some Measures of Central Tendency and Dispersion

Sort the data in ascending order first:

$$x_i \ , \ i = 1, \ldots, \ n \qquad x_1 \le x_2 \ldots \le x_n$$

Central Tendency – Median

For *n* odd: Median = $x_j \ , \quad j = (n+1)/2$

For *n* even: Median = $\dfrac{x_j + x_k}{2} \ , \quad j = \dfrac{n}{2}, \quad k = \dfrac{n}{2} + 1$

Dispersion – Range (smallest and largest values) $\left[x_1, \quad x_n \right]$

Dispersion – Percentile Ranges

How would you calculate the 25th and 75th percentiles of the data?

Calculating Percentiles from Sample Data

p^{th} percentile of a data set is the smallest number x such that p% of the data is less than x

1. Sort the data in ascending order:

$x_i, i = 1, \ldots, n \quad x_1 \leq x_2 \ldots \leq x_n$

$1 \quad 2 \ldots \quad n$

2. Calulate rank or index $== \dfrac{p(n-1)}{100} + 1 = I_R + F_R$

Integer part of answer = I_R = integer rank; Fraction part of answer = F_R

3. Calculate p^{th} percentile by interpolation:

Find observations with ranks I_R & $I_R + 1$ $\left(x_{I_R} \ \& \ x_{I_R+1}\right)$

$$p^{th} \text{ percentile} = x_{I_R} + F_R \cdot \left(x_{I_R+1} - x_{I_R}\right)$$

Example of Calculating Percentiles from Sample Data

Find 30th percentile of sample data (already sorted)
3, 7, 8, 9, 14

p = percentile = 30
n = number of data points in sample = 5

$$\text{rank or index} = \frac{p(n-1)}{100} + 1 = \frac{30(5-1)}{100} + 1 = 2.2$$

$$I_R = 2 \qquad F_R = 0.2$$

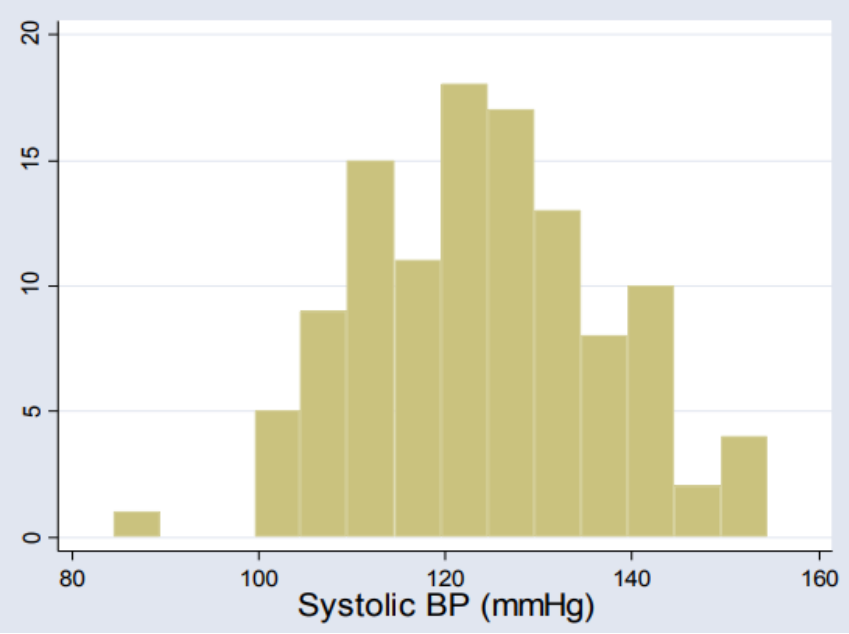
$$x_{I_R} = x_2 = 7$$

$$x_{I_R+1} = x_3 = 8$$

$$30^{th} \text{ percentile} = x_2 + F_R \cdot (x_3 - x_2) = 7 + 0.2 \cdot (8-7) = 7.2$$

Data Display– Summarizing/Visualizing Data

Histograms are used to visually summarize data: divide all the possible values of x into bins and count the number of observations that fall into each bin. Plot the count (frequency) in each bin versus the midpoint of the bin.



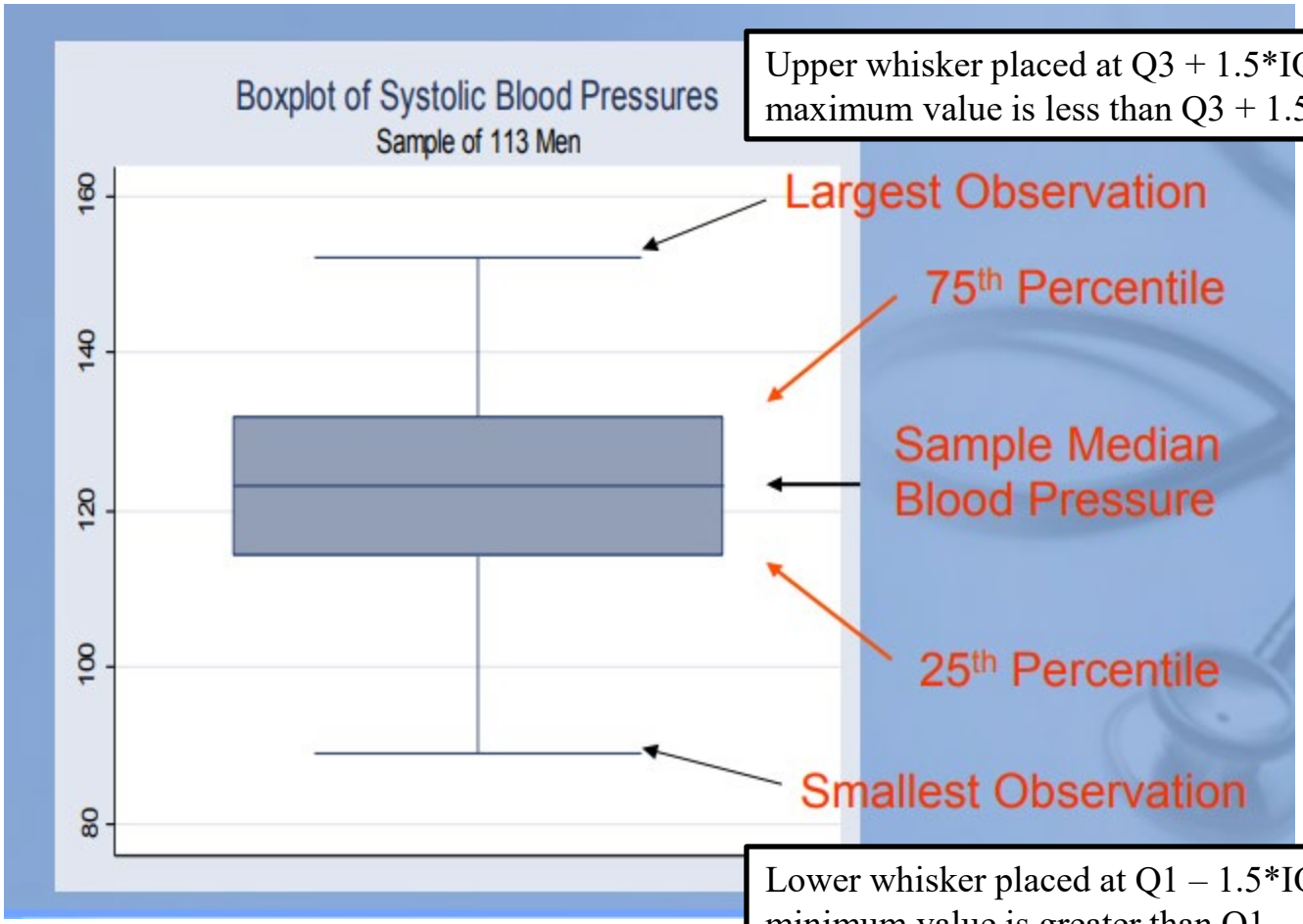
Data from the CTSI
Penn State Univ
$n = 113$

Data can be summarized by their 25th, 50th (median) and 75th percentiles (which are the Q1, Q2, and Q3 **quartiles**, respectively). The difference between the 75th and 25th percentiles is referred to as the **interquartile range (IQR)**, which is also a measure of dispersion.

IQR of Systolic BP = 132 – 116 = 16 mmHg

Data Display– Summarizing/Visualizing Data

Box plots are a common way to visually show the quartiles and other summarizing information



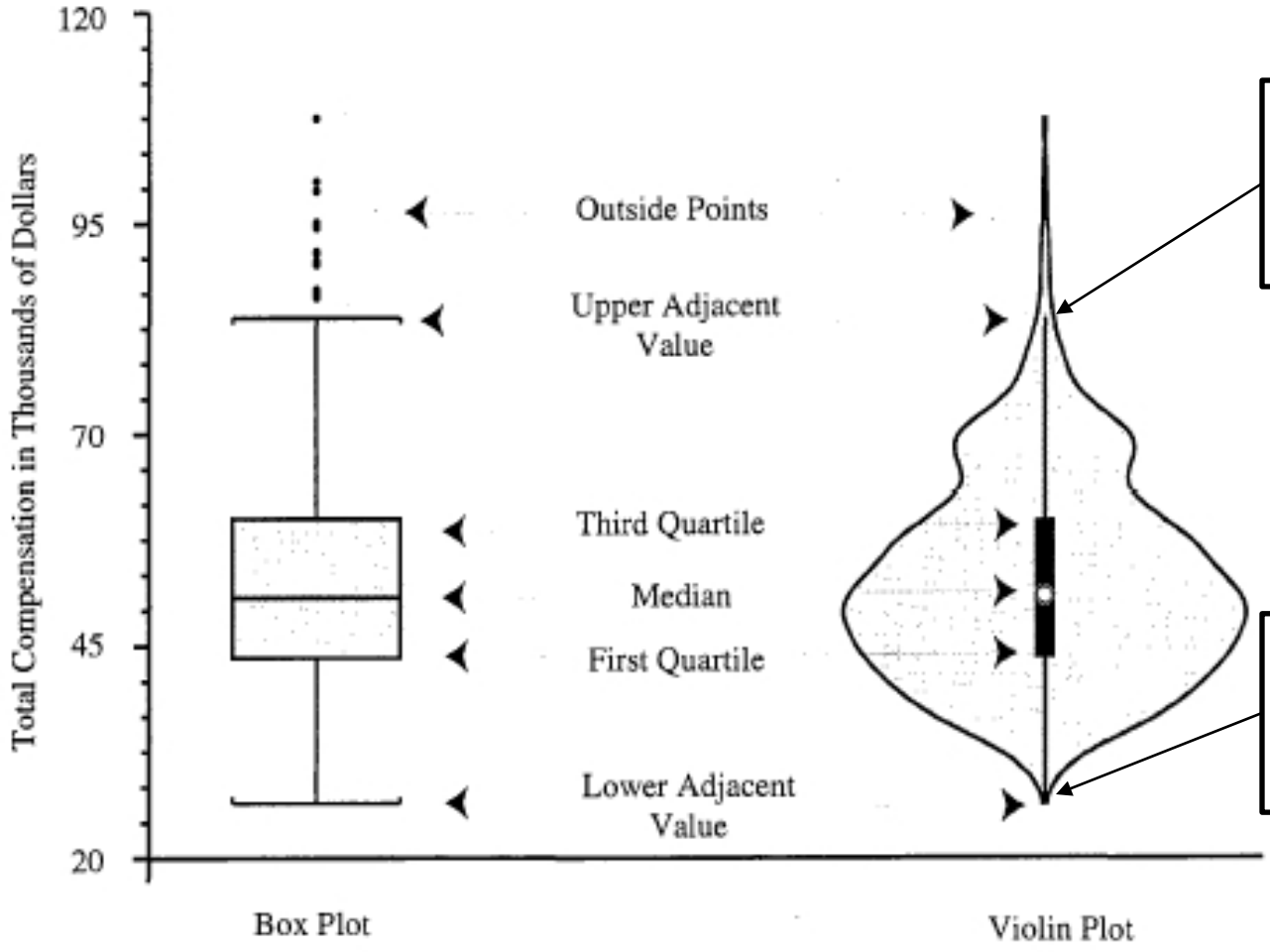
Upper whisker placed at Q3 + 1.5*IQR (or maximum value, if maximum value is less than Q3 + 1.5*IQR)

Lower whisker placed at Q1 – 1.5*IQR (or minimum value, if minimum value is greater than Q1 – 1.5*IQR)

There are many variations for displaying data using box plots

Data Display– Summarizing/Visualizing Data

Violin plots are a common way to visually show the quartiles and other summarizing information, as well as the distribution of individual values



Upper adjacent value extends to Q3 + 1.5*IQR (or maximum value, if maximum value is less than Q3 + 1.5*IQR)

Lower adjacent value extends to Q1 – 1.5*IQR (or minimum value, if minimum value is greater than Q1 – 1.5*IQR)

There are many variations for displaying data using violin plots

Data Display– Summarizing/Visualizing Data

Qualitative assessment of whether data follow a Normal distribution can be done by checking some of the following, depending on the size of the data set:

~2.5% of the values of X should be below the value $\bar{x} - 2s$

~16% of the values of X should be below the value $\bar{x} - s$

~50% of the values of X should be below the value $\bar{x}$

~84% of the values of X should be below the value $\bar{x} + s$

~97.5% of the values of X should be below the value $\bar{x} + 2s$

~68% of the values of X should be between $\bar{x} - s$ & $\bar{x} + s$

~95% of the values of X should be between $\bar{x} - 2s$ & $\bar{x} + 2s$

In later lectures we will follow up on these ideas using plots, as well as a formal statistical test

How Good are these Estimates?

$\bar{x}$ is an estimate of μ $\quad$ s is an estimate of σ

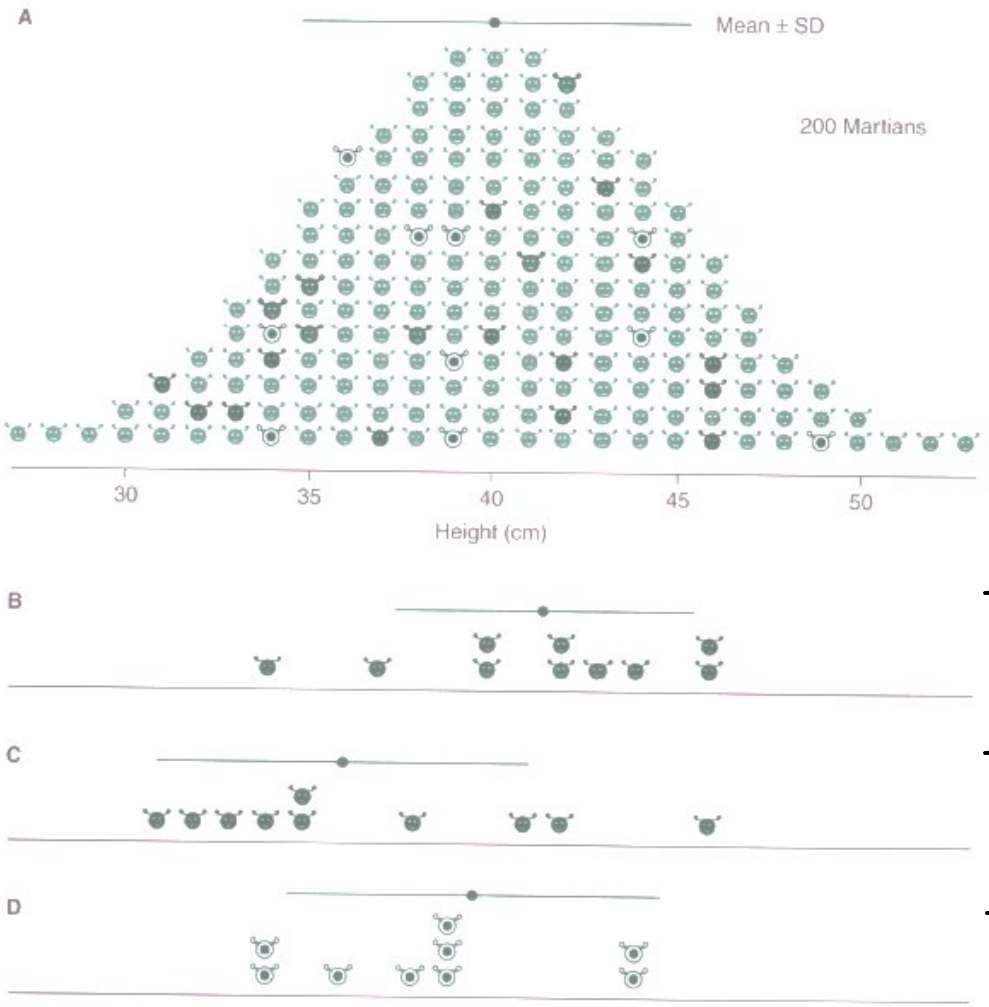


FIGURE 2-5. If one draws three different samples of 10 members each from a single population, one will obtain three different estimates of the mean and standard deviation.

$\mu = 40.$ cm

$\sigma = 5.0$ cm

$n = 10$

$\bar{x}_1 = 41.5$

$s_1 = 3.8$

$\bar{x}_2 = 36.$

$s_2 = 5.$

$\bar{x}_3 = 40.$

$s_3 = 5.$

• • •

Distribution of Sample Mean Values of Size $n = 10$

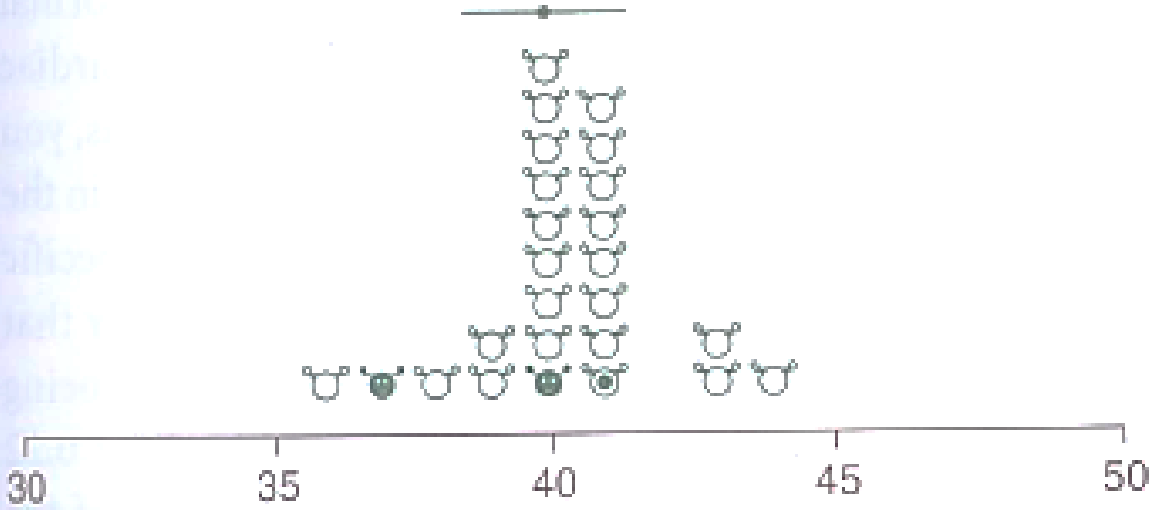
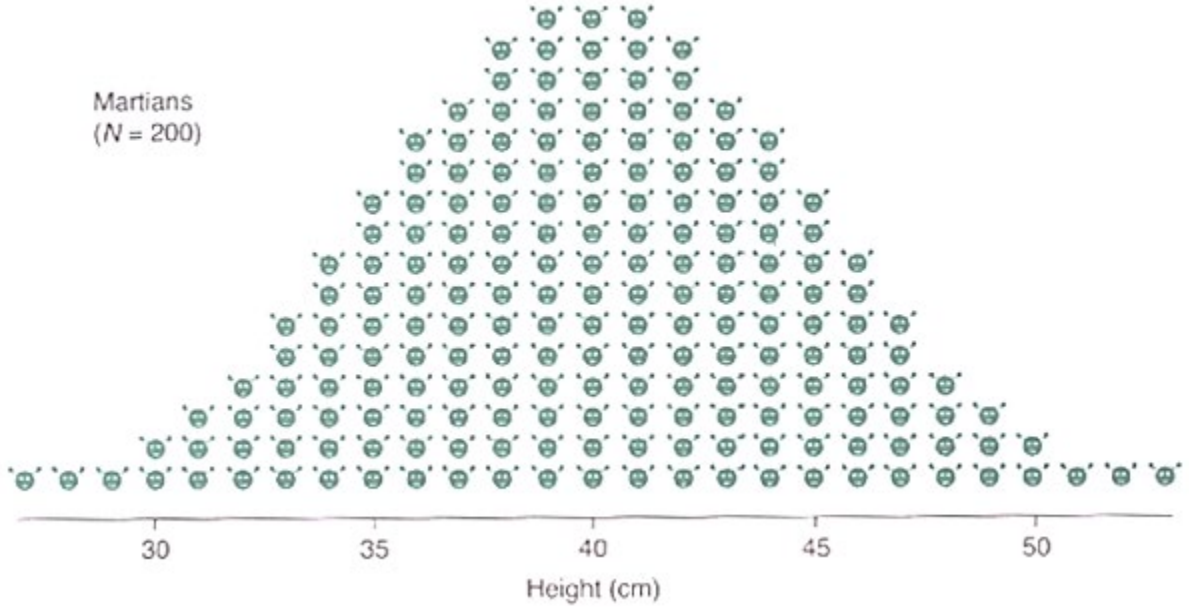


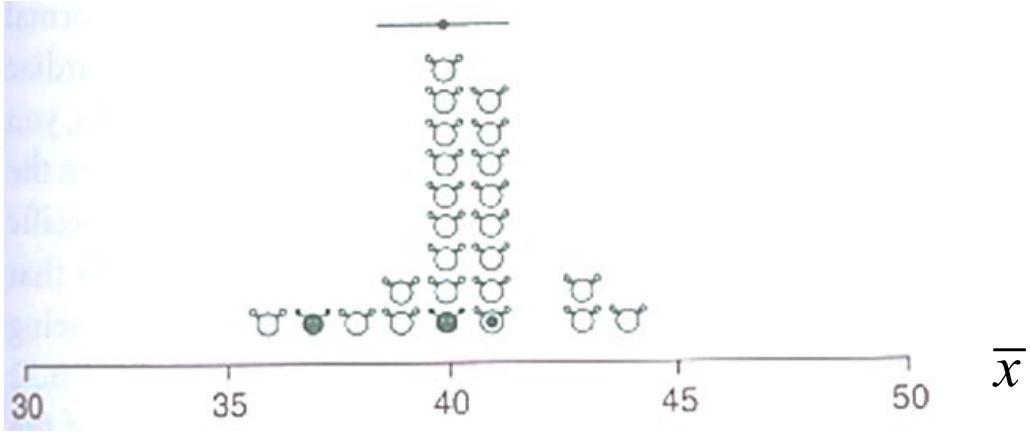
FIGURE 2-6. If one draws more and more samples—each with 10 members—from a single population, one eventually obtains the population of all possible sample means. This figure illustrates the means of 25 samples of 10 Martians each drawn from the population of 200 Martians shown in Figures 2-1 and 2-5A. The means of the three specific samples shown in Figure 2-5 are shown using corresponding symbols. This new population of all possible sample means will be normally distributed regardless of the nature of the original population; its mean will equal the mean of the original population; its standard deviation is called the standard error of the mean.



For Height

$\mu = 40.$ cm

$\sigma = 5.0$ cm



Mean Height (cm)

Sample mean of size 10

Sample mean of a sample of size n, called $\bar{x}$, is now a new random variable

Mean of sample means: $\mu_{\bar{x}}$

Standard deviation of sample means: $\sigma_{\bar{x}}$
(also called SEM: standard error of the mean)

$$\mu_{\bar{x}} = \mu$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

← Central Limit Theorem

Idea of Central Limit Theorem:

- If a population has mean μ and variance σ^2
- If random samples of size $n \geq 30$ (with replacement) are taken from population
- Sampling distribution of sample means ($\bar{x}$) is approximately Normally distributed with $\mu_{\bar{x}} = \mu$ and $(\sigma_{\bar{x}})^2 = \sigma^2/n$, even if the underlying population is not Normally distributed

But how can we estimate $\sigma_{\bar{x}}$ from a single sample of size n?

Recall $\sigma_{\bar{x}} = \dfrac{\sigma}{\sqrt{n}}$

Best estimate of $\sigma_{\bar{x}}$ from a single sample of size n is

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

In Summary

Given a sample of size n: $x_i \ , \ i = 1, \ldots, \ n$

Sample mean: $\bar{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$ Sample variance: $s^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2$

Standard error of mean: $s_{\bar{x}} = \dfrac{s}{\sqrt{n}}$ Standard error of variance: $s_{s^2} = s^2\sqrt{\dfrac{2}{n-1}}$

III. Classification of Data

Interval Data (measurement scale has constant intervals)

- Continuous (can take on any value within a range)
  - e.g., weight, pH, LDL, blood pressure, height
- Discrete (limited to a set of values)
  - e.g., # tender joints, # asthmatic episodes

Categorical Data (measurement scale is set of categories)

- Ordinal (natural ordering to the data)
  - e.g., cancer stage (I,II,III,IV),
    pain level (none, mild, moderate, strong, severe)
- Nominal (unordered scale)
  - e.g., sex (F, M), cause of death (accident, heart disease, etc.)

IV. BME 423 Student Data Summary

1 Q1. Eye Color

1 Q2. Handedness

1 Q3. Estimated Length of Left Index Finger (nearest cm)

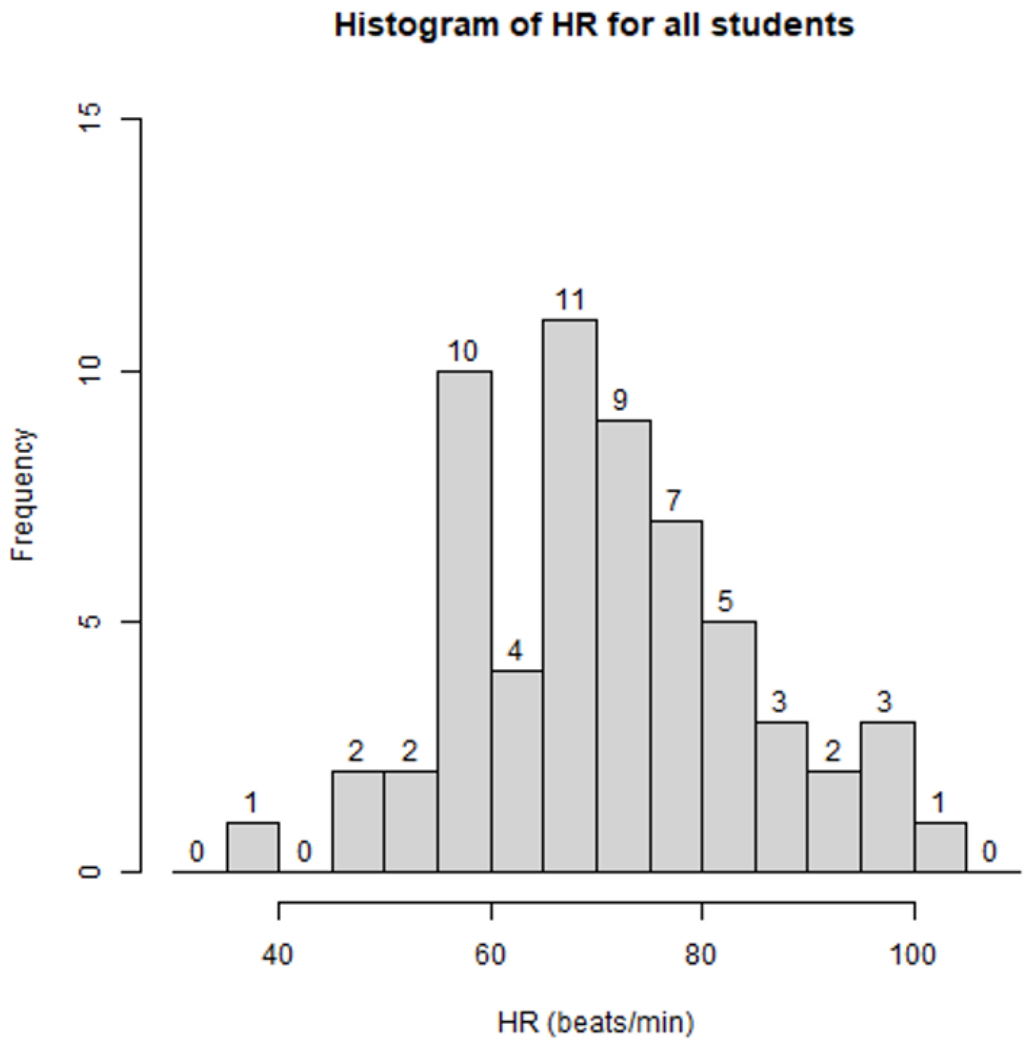
1 Q4. Do you Prefer Dogs or Cats?

1 Q5. Class Level

1 Q6. Height (inches)

1 Q7. Heart Rate (beats per minute)

Histogram of HR for All Students in BME 423 – The Population



N = 60

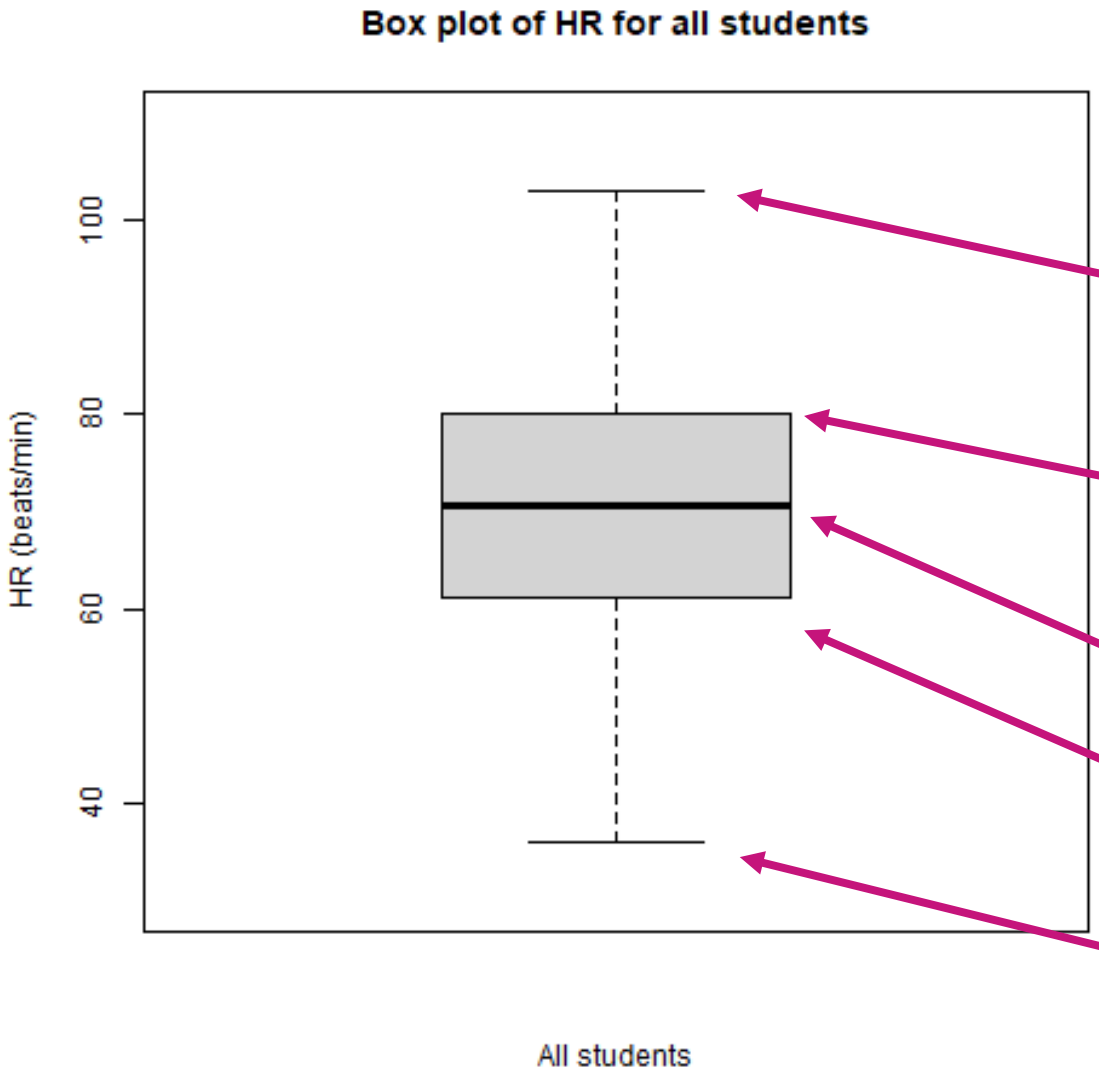
μ = 72.1 beats/min

σ = 13.7 beats/min

```
       HR
Min.   : 36.00
1st Qu.: 61.50
Median : 70.50
Mean   : 72.07
3rd Qu.: 80.00
Max.   :103.00
```

IQR = 80 - 61.5 = 18.5 beats/min

Box Plot of HR for All Students



max = 103 beats/min

75% = 80 beats/min (3rd quartile)

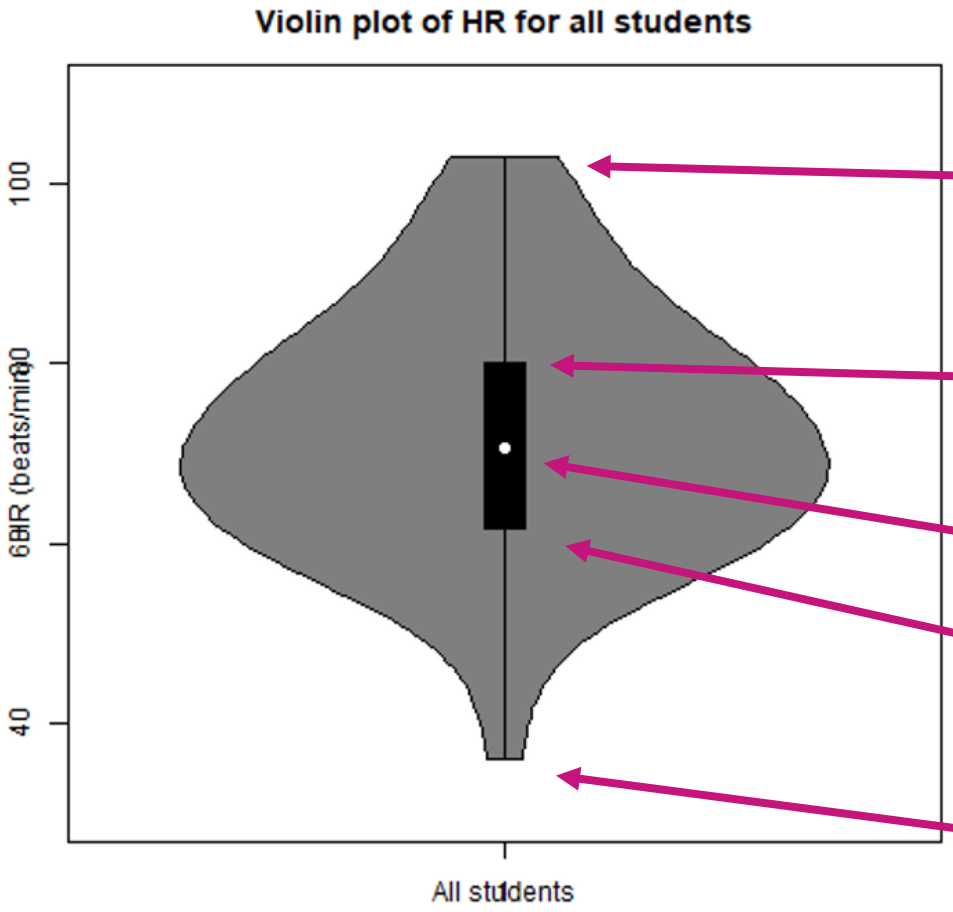
median = 70.5 beats/min (50%)

25% = 61.5 beats/min (1st quartile)

min = 36 beats/min

IQR = 80 - 61.5 = 18.5 beats/min

Violin Plot of HR for All Students



max = 103 beats/min

75% = 80 beats/min (3rd quartile)

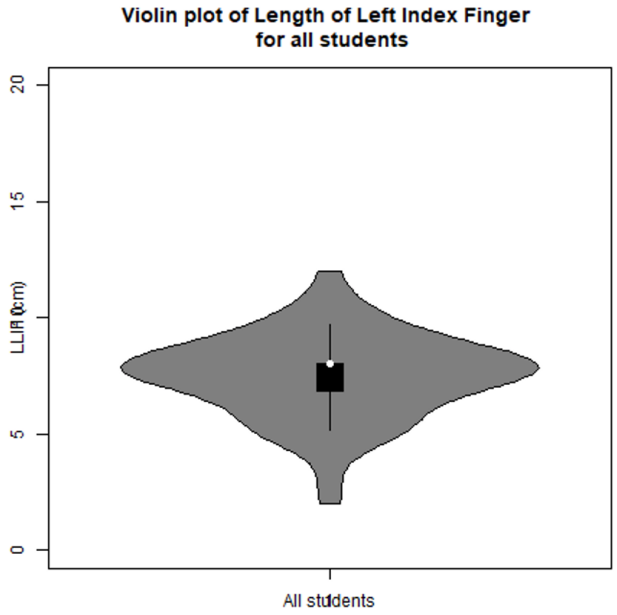
median = 70.5 beats/min (50%)

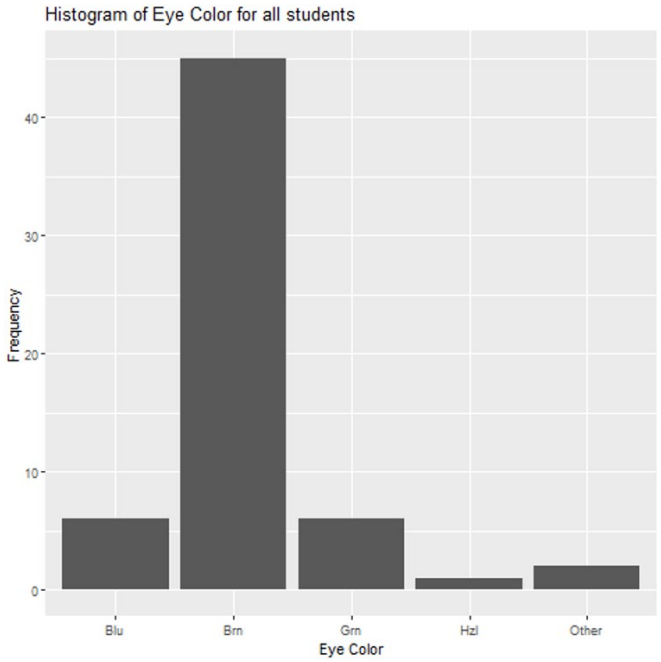
25% = 61.5 beats/min (1st quartile)

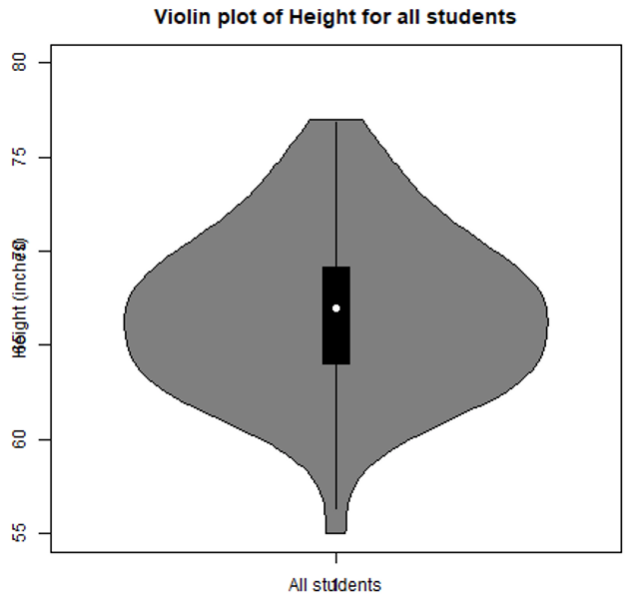
min = 36 beats/min

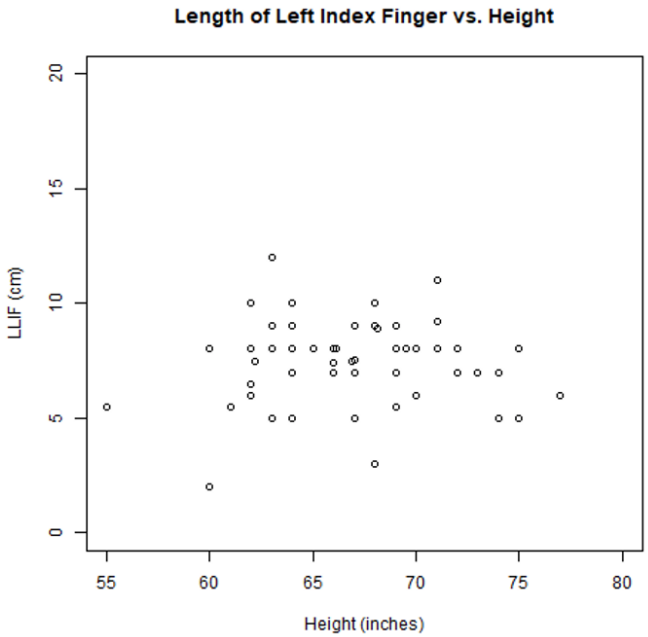
IQR = 80 - 61.5 = 18.5 beats/min

Some Other Results for All Students

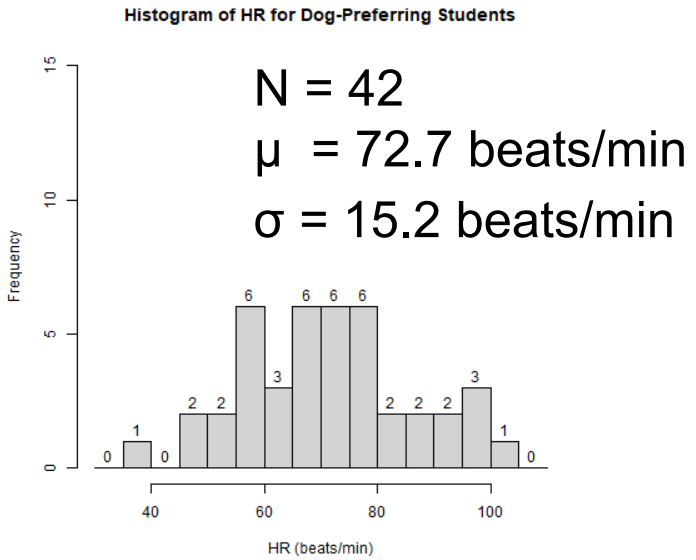




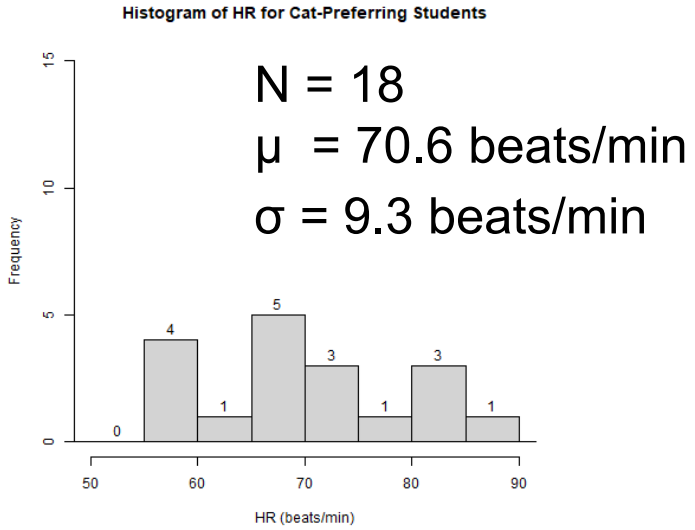




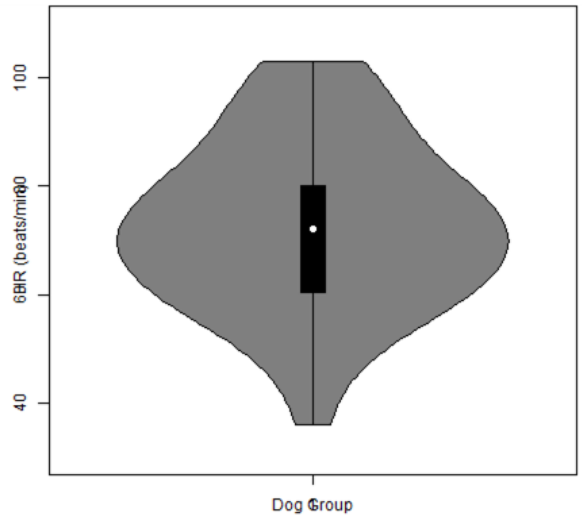
Histogram of HR for Dog-Preferring Group



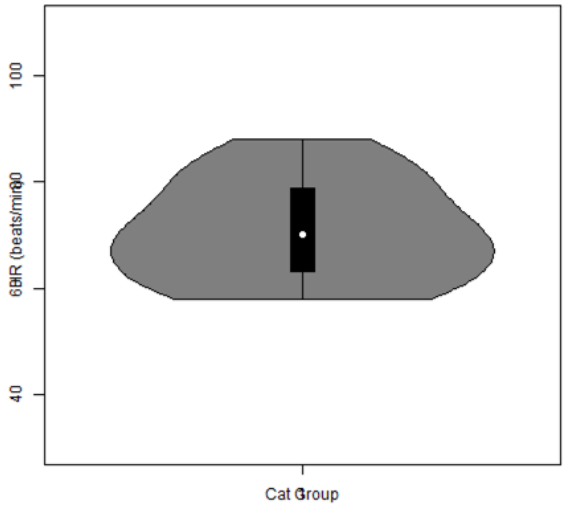
Histogram of HR for Cat-Preferring Group



Violin plot of HR for Dog Group



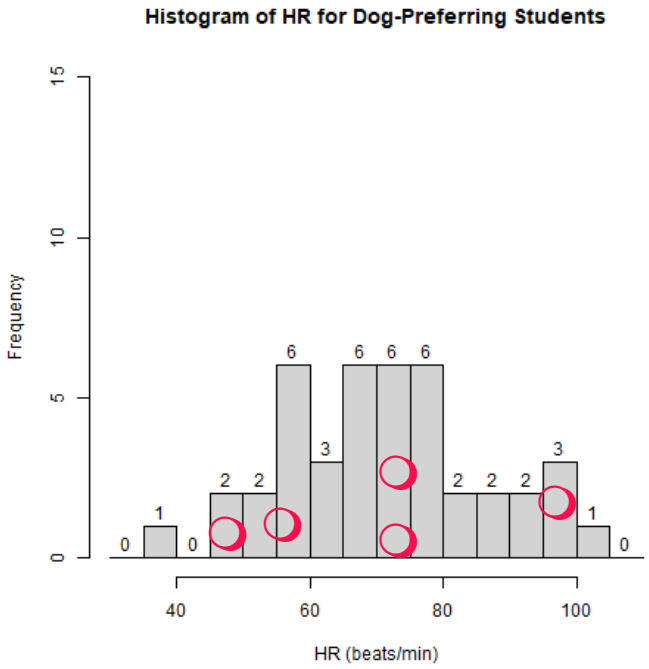
Violin plot of HR for Cat Group



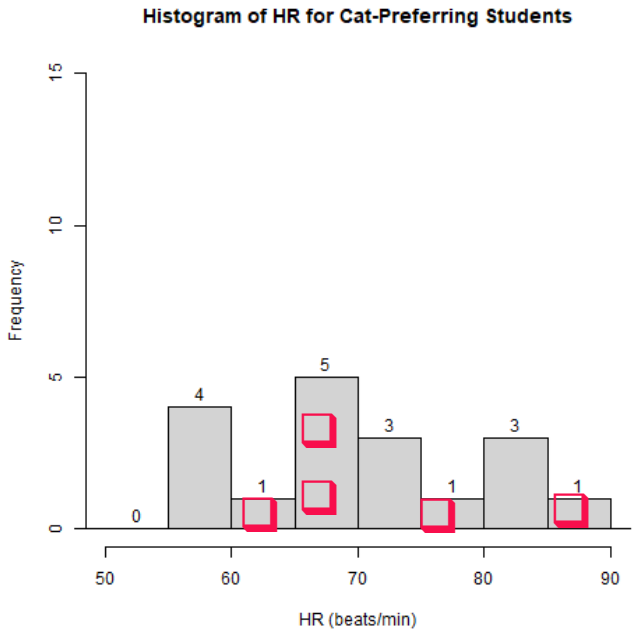
Type of question we will address:

Based on the HR from a sample from the Dog Group and a sample from the Cat Group ($n = 5$ in the example below), is there a difference in HR of BME 423 students who prefer dogs versus those who prefer cats?

Histogram of HR for Dog Group



Histogram of HR for Cat Group



We will need a statistical test to make an inference from these samples with regards to the underlying populations; this statistical test will need to indicate the probability that the inference is correct